

Frozen Base + Strategy Text

Prefix-only (B3)

LoRA + Strategy Text

Warm-start (B7)

Prefix→LoRA (B9)

Prefix+LoRA (B4)

B6 contrastive

B6 contrastive (no orth)

B6 orth=1.0

B6 cls=5.0

B6 gradient routing

+ Orth (B5)

B6 fix (orth+cls)

Random (11.1%)

PPL —

PPL 10.54

PPL 7.81

PPL 8.18

PPL 8.07

PPL 7.93

PPL 8.06

PPL 7.90

PPL 8.14

PPL 7.92

PPL 7.91

PPL 7.80

PPL 7.97

Prefix

Text

LoRA

Joint / Staged

Diagnostic

Strategy Accuracy (%)

0 10 20 30 40 50 60